



MEGN 300: Spring 2026

Design Challenge Two: DTMF Decoder

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Problem Introduction:

Dual-Tone Multi-Frequency (DTMF) systems are primarily used by analog telephones to encode and decode the audio signals created when a specific key is pressed on the keypad [1]. Every key on the keypad corresponds to a combination of two frequencies, and these frequencies are used to denote a specific index value in the table of unique frequencies. The receiving end of the system decodes the two frequencies created and finds the corresponding number. This is why when 7 is pressed for example, a low frequency of 852 Hz and high frequency of 1209 Hz are created allowing the system to quickly recognize the specific number. The simplicity of this system revolutionized communication, replacing traditional dial systems with the much more efficient touch tone process which allowed for automated menu navigation and permitted other interactions that were previously impossible [1]. As a team, we are tasked with creating a DTMF decoder utilizing digital analog filters that automatically and accurately outputs the correct keypad number when a DTMF signal is received.

System Overview:

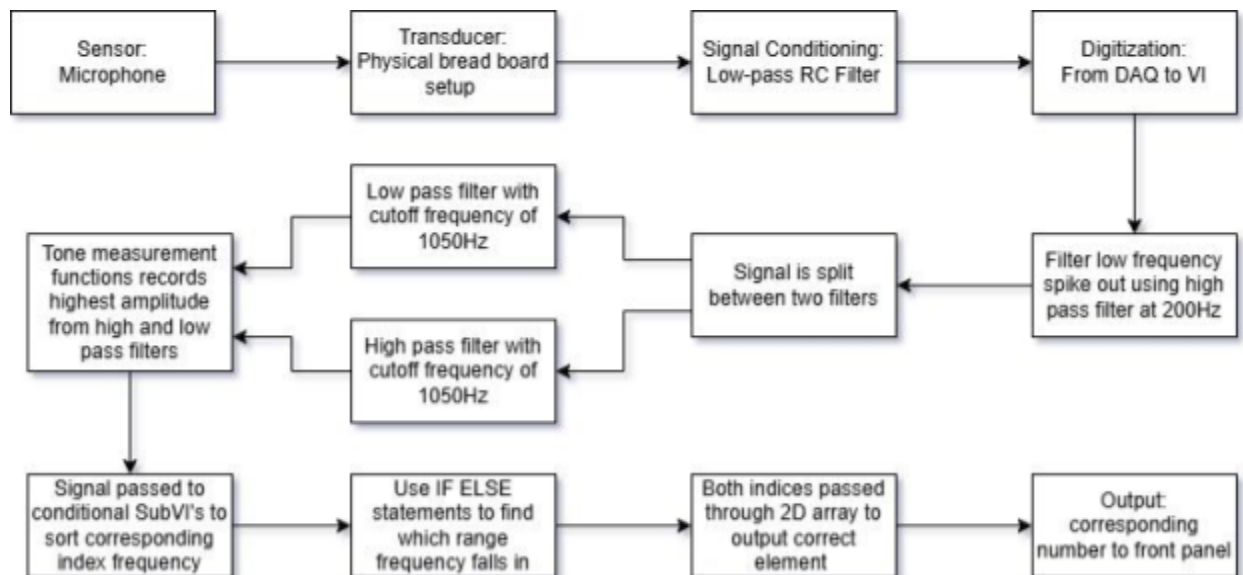


Figure 1: System Flow Diagram

The flow diagram depicted above in Figure 1 shows the flow of information through the hardware and software of our DTMF decoder. The diaphragm in the microphone picks up any and all audio signals in the environment, and our circuit captures the voltage change that such audio signals induce and passes them through a low-pass RC filter, which removes all frequencies above our cutoff frequency to prevent aliasing. This signal is then passed into the DAQ to be digitized, after which it enters our VI to be processed.

The signal enters an additional high pass filter to remove any large low frequencies below the DTMF range, and is then split into two separate filters: a low pass filter and an additional high

pass filter, both with a cutoff frequency of 1050 Hz in order to separate the high and low tones in a DTMF signal. These two tones are measured by two tone measurement functions, which both produce a single frequency scalar.

Finally, both tones are passed through an IF-ELSE chain to determine the number that they correspond to. This number is then pushed to an indicator in the front panel of the VI as the final output.

Sample Rate Determination:

To determine an appropriate sample rate, we first established a max cutoff frequency of 1500 Hz, which sits neatly above the highest frequency tone used for DTMF. As is demonstrated below in the calculation for the RC value of the anti-aliasing filter, this max cutoff frequency yields a Nyquist frequency of 1591 Hz to account for the real-world resistance and capacitance. To prevent aliasing, we set our sample rate to twice the value for the nyquist frequency: 3183 Hz.

Anti-Aliasing Filter:

The following equation relates cutoff frequency to the resistance and capacitance used in an analog RC filter:

$$f_c = \frac{1}{2\pi RC}$$

Isolating RC and substituting cutoff frequency for 1500 Hz (the previously determined max frequency) yields the following value for RC:

$$RC = \frac{1}{2\pi f_c} = \frac{1}{2\pi(1500)} = 0.1061 \text{ ms}$$

To achieve an approximate RC value with the available components, a 10kΩ resistor and a 10 nF capacitor were selected (yielding $RC = 0.1 \text{ ms}$) and placed in series with the output signal of the provided microphone module, as shown below in Figure 2.

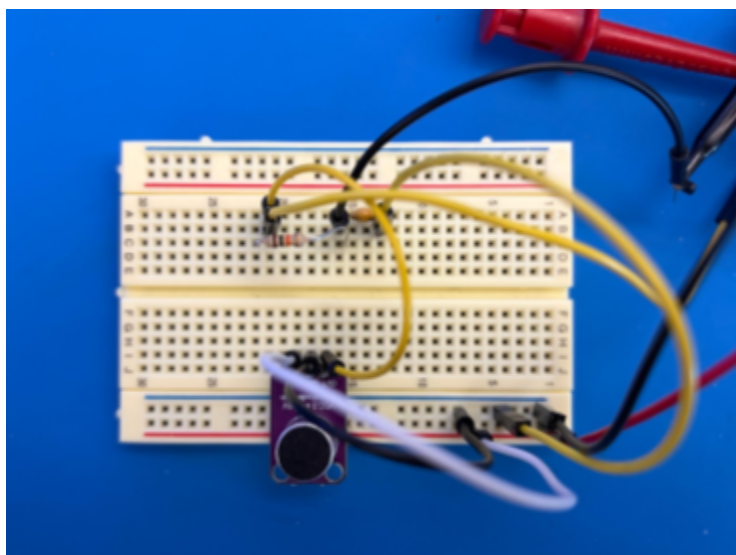


Figure 2: Analog Low Pass Filter

Two leads were placed across the resistor in the circuit to properly filter low frequencies through, creating an effective low pass filter to remove high frequencies that may contribute to aliasing.

System Documentation:

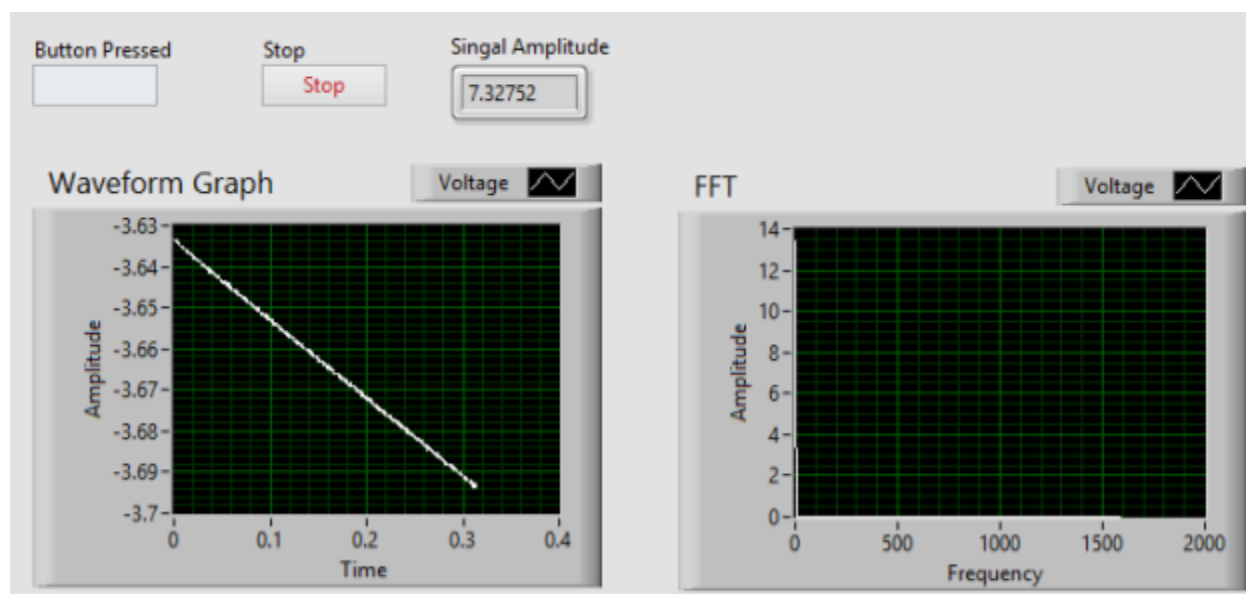


Figure 3: LabView VI Front Panel

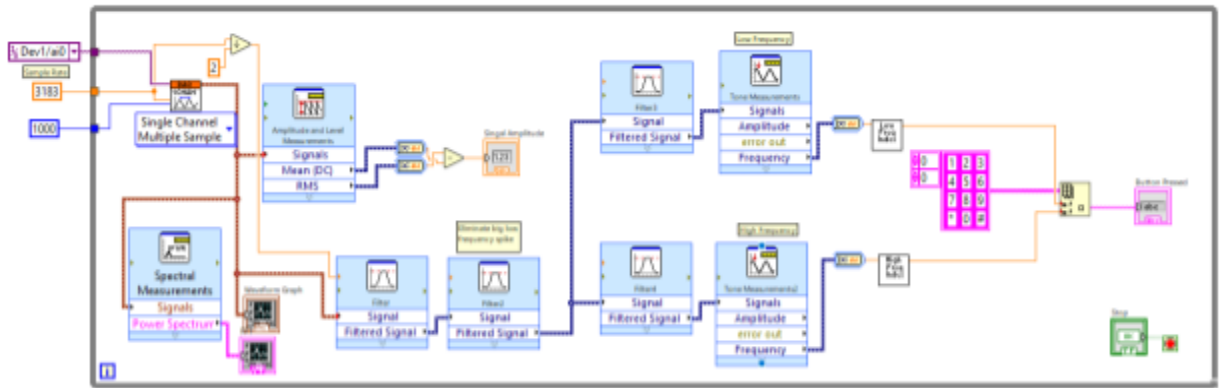


Figure 4: LabView VI Block Diagram

Figures 3 and 4 show the front panel and block diagram running the VI. This combines with the hardware shown previously in Figure 2 to provide an accurate determination of the DTMF tone being played near the microphone. The signal travels from the sample measurement block and splits into four paths: one into a spectral measurement block to produce an FFT chart in which each prominent frequency can be observed, another into a waveform graph to show the raw signal, a third into an amplitude and level measurements block to gather data for the SNR value, and a fourth into a series of filters to process and separate the individual frequencies within the DTMF tone.

After the high and low frequencies are measured by their respective tone measurement blocks, they are then passed into two SubVIs to determine what range the frequencies fall into (pictured below in Figures 5 and 6), after which they are assigned an index corresponding to the correct DTMF keypad element.

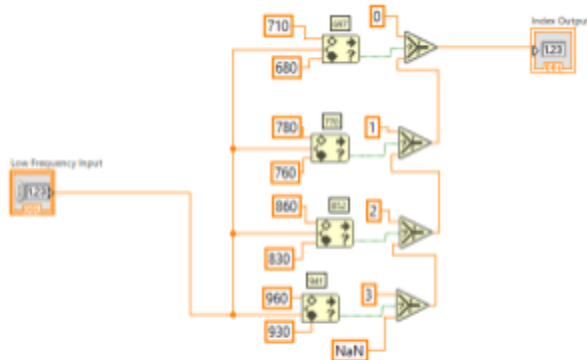


Figure 5: Low Frequency Index SubVI

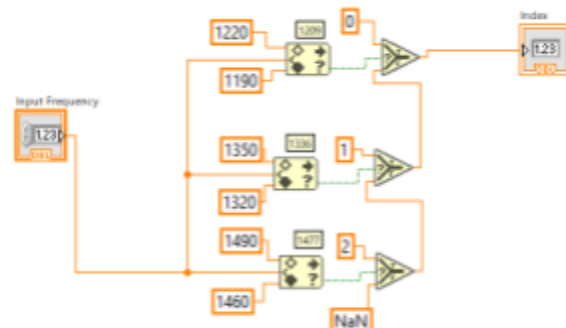


Figure 6: High Frequency Index SubVI

Tone Determination:

For determining the specific tone we estimated a reasonable tolerance given the frequencies in the project description. We decided on a tolerance of approximately ± 10 Hz to ensure the correct DTMF low and high frequency signals could be identified precisely (specific ranges are shown above in Figures 5 and 6), leaving some space for variance in speaker quality and noise.

This tolerance proved to be acceptable and allowed for precise determination of the tone every time. [This video](#) demonstrates the functionality of the decoder.

Discussion:

As demonstrated in the linked video above, the system does an excellent job of determining the keypad symbol being pressed using the DTMF tone. To determine if the system could withstand high frequencies above the cutoff frequency, an additional frequency above 2000 Hz was played at the same time as a valid DTMF tone. The system performed well with the anti-aliasing filter until the 2000 Hz frequency was being played much closer to the microphone than the DTMF tone, at which point the system could no longer recognize the tone being played. Without the anti-aliasing filter, the system performed much worse. While it could still recognize the low frequency within the DTMF tone, the high frequency was being overridden by the 2000 Hz tone, which had a greater signal amplitude.

Signal to Noise Ratio:

SNR is used to determine how noisy a specific waveform is.

$$SNR = \frac{\text{Signal Amplitude}}{\text{Noise Amplitude}}$$

For our signal amplitude we used #1 on the keypad as our control variable. The Labview VI automatically calculates the RMS and subtracts the mean value from the RMS value recorded.

Signal Amplitude Calculation:

Test 1: 0.243586

Test 2: 0.237323

Test 3: 0.270579

Test 4: 0.242756

Test 5: 0.253738

Avg = 0.2495964

Noise amplitude was found by recording the highest amplitude of ambient noise when running the VI for a period of time.

Noise Amplitude Calculation:

Test 1: 0.220042

Test 2: 0.220291

Test 3: 0.212178

Test 4: 0.208153

Test 5: 0.211597

Avg = 0.210976

$$SNR = \frac{0.2495964}{0.210976} = 1.1831$$

1.1831 > 1 which means the signal can be distinguished from the noise.

Appendix:

[1] "Understanding DTMF: An Introduction to Dual-Tone Multi-Frequency Technology," Apr. 2026. [Online]. Available: [[Understanding DTMF: An Introduction to Dual-Tone Multi-Frequency Technology - \[Updated April 2026 \]](#)]. Accessed: Apr. 8, 2026.